Bayesian adaptive distributed lag models

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Abstract

Distributed lag models (DLMs) express the cumulative and delayed dependence between pairs of time-indexed response and explanatory variables. In practical application, users of DLMs examine the estimated influence of a series of lagged covariates to assess patterns of dependence. Much recent methodological work has sought to develop flexible parameterisations for smoothing the associated lag parameters that avoid overfitting. However, this paper finds that some widely-used DLMs introduce bias in the estimated lag influence, and are sensitive to the maximum lag which is typically chosen in advance of model fitting. Simulations show that bias and misspecification are dramatically reduced by generalising the smoothing model to allow varying penalisation of the lag influence estimates. The resulting model is shown to have substantially fewer effective parameters and lower bias, providing the user with confidence that the estimates are robust to prior model choice. An efficient inference scheme is described in which the lag coefficients are integrated out analytically and the remaining low-dimensional posterior is sampled with the No-U-Turn Sampler, with a cost per likelihood evaluation that does not grow with the length of the series.

Keywords: Distributed lag model; penalized spline; adaptive smoothing

1 Introduction

Distributed lag models (DLMs) find application whenever the influence of a time-indexed independent variable is delayed and spread over time. Although DLMs are widely applicable, their development was originally motivated by problems in econometrics [Nerlove, 1958, Almon, 1965, Zellner and Geisel, 1970, Haugh and Box, 1977] and they have more recently experienced a surge in popularity in environmental epidemiology [Zanobetti et al., 2000, Welty et al., 2009, Gasparrini et al., 2010]. Crucially, DLMs enable direct interpretation of the influence of a temporal exposure ensemble, which is particularly useful for characterising the total health impact of persistent environmental exposures such as air pollution or temperature [Wyzga, 1978, Gasparrini et al., 2010]. Lag models can help to identify subtle types of time-dependence such as 'mortality displacement', which occurs when exposure related mortality diminishes a vulnerable subpopulation, resulting in lower mortality in subsequent time intervals. Mortality displacement has been widely documented

[Schwartz, 2000, Braga et al., 2001] and is conspicuous where lag influence is estimated to have a protective effect at low or moderate lags for what are otherwise harmful exposures [Zanobetti et al., 2000]. Correctly identifying these effects depends on obtaining unbiased estimates of the underlying lag influence, and it is therefore essential that the DLM is correctly specified. Since no previous study has formally quantified the estimation cost of model misspecification, this paper provides a comparison of various existing techniques.

If a time-varying explanatory variable X_t has some influence on a response variable Y_t over a set of lags $0, \ldots, p$, then a simple DLM can be expressed as the regression model $Y_t = \sum_{i=0}^{p} X_{t-i}\beta_i + \epsilon$, where ϵ is an independent error process and the parameters $\boldsymbol{\beta} = (\beta_0, \beta_1, \ldots, \beta_p)$ are referred to as the 'lag curve' or lag function, as a function of i. $\boldsymbol{\beta}$ cannot be reliably estimated using ordinary generalised linear models when X_t and X_{t-1} are correlated, and regularisation using shape constraints or smoothness restrictions over the lag parameters $\boldsymbol{\beta}$ are frequently used. Recent studies have implemented various forms of flexible parameterisation using polynomial splines and smoothness penalties, for example by combining uniformly spaced B-splines with ridge penalisation [Zanobetti et al., 2000], logarithmically spaced B-splines [Gasparrini et al., 2010], penalised B-splines [Rushworth et al., 2013, Obermeier et al., 2015], and Bayesian penalised splines with varying ridge penalties [Muggeo, 2008]. Spline basis representations ensure a minimum level of smoothness, and offer computational relief when there are fewer knots than lags, which reduces the total number of parameters to estimate. Although smoothness restrictions primarily serve to stabilise the model, they can also be exploited to incorporate additional structure over the shape of the estimated lag curve. For example, it is widely assumed that the true lag curve has more curvature at short lags than at large lags, reflecting the intuition that recent events are generally the most influential, and influence should decay rapidly to zero for increasing lags. Further structure can also be used to avoid discontinuity between the influence of the final modelled lag $\beta_p \neq 0$ and the implicitly assumed $\beta_{p+j} = 0$. Some examples include Koyck [1954] who restricted $\boldsymbol{\beta}$ to decay geometrically with increasing p; Muggeo [2008] who combined a B-spline basis with parameters which were shrunk by a lag-varying ridge penalty; Gasparrini et al. [2010] used logarithmic knot placement in the lag dimension to enable varying smoothness; and Welty et al. [2009] constructed a Gaussian process prior distribution that enforces smaller covariances on lag parameters at short lags than longer ones. While each of these approaches achieves a level of lag-varying smoothness, each incorporates

a parametric assumption on the type of penalisation that occurs, which may not be appropriate in general.

A further problem arises because the maximum lag p must be chosen before model fitting: this is sometimes justified by drawing on previously published studies, for example, in several studies of the delayed effects of temperature on human health, $20 \leq p \leq 30$ is consistently reported [Braga et al., 2002, Armstrong, 2006, Gasparrini and Armstrong, 2013]. In general, there is a cost to getting p wrong: too large and the model risks overfitting, and too small can bias the resulting lag function estimate. Few studies have investigated sensitivity of different models to choices of p, although several authors recognise that this is an ongoing difficulty for DLMs. Heaton and Peng [2012] and Heaton and Peng [2014] provide notable exceptions, and estimate p as part of a two-stage approach. However, DLMs are often applied in settings where the true lag curve is believed to decay gradually to 0, and in such settings, p is likely to be poorly estimated and will lack a meaningful interpretation.

This paper proposes a pragmatic strategy that ensures flexible and automatic smoothing over the lag structure that avoids the need to either estimate or choose p in advance. By exploiting the property that very large lags have close to 0 influence, an automatic lag-dependent smoother is constructed that ensures strong penalisation at high lags so that p can be made arbitrarily large without the risk of overfitting. This follows the tradition of richly specifying a hierarchical model and ensuring parsimony by liberal use of regularisation and smoothing. A new DLM is described in Section 2, its computation in Section 3, and in Section 4 a simulation study illustrates the DLM's desirable properties, especially in comparison to existing approaches. This study compares bias resulting from getting p wrong, which is itself a central contribution, as DLMs have not previously been compared in this manner. Section 5 applies the model to four real data sets: daily stock market returns, intraday order flow and prices at one-second resolution, air pollution and mortality in Chicago, and COVID-19 cases and deaths. Section 6 concludes the paper with an overview of the main findings and a discussion of the implications for users of distributed lag models.

2 Adaptive distributed lag models

Define $\mathbf{y} = (y_1, \ldots, y_n)$, a response variable time series of length n, and $\mathbf{x} = (x_1, \ldots, x_n)$, a time series of an exposure or input, whose influence on $\mathbf{y}$ up to some lag p is to be estimated. A Gaussian DLM for these data can be written as $y_i \sim \mathrm{N}\left(\sum_{j=0}^{p} x_{i-j}\beta_j,\ \sigma^2\right)$ where $i = p+1, p+2, \ldots, n$, and σ^2 is the error variance. The lag parameters $(\beta_0, \beta_1, \ldots, \beta_p)$ could be penalised and estimated directly, but for ease of comparison with existing approaches, these will be projected onto a B-spline basis of K functions uniformly spaced over the lag range $0, \ldots, p$. The basis representation can be expressed as $\beta_j = \sum_{k=1}^{K} B_k(j) b_k$ where the further set of parameters $\mathbf{b} = (b_1, \ldots, b_K)$ determine the fitted lag curve. The model can be expressed as $\mathbf{y} \sim \mathrm{N}(\mathbf{Xb},\ \sigma^2\mathbf{I})$, where $\mathbf{X}_{ik} = \sum_{j=0}^{p} x_{i-j} B_k(j)$. To control the roughness of b_k in a way that enables variable smoothing, a similar approach to Brewer and Nolan [2007] and Reich and Hodges [2008] is used, in which the intrinsic autoregressive (ICAR) prior distribution of Besag et al. [1991] is generalised. The conditional prior distribution for $\mathbf{b}$ under the ordinary ICAR is $\mathbf{b} \sim \mathrm{N}(\mathbf{0}, \frac{1}{\lambda}\mathbf{P}^-)$, resulting in a conditional prior density of

$$\pi(\mathbf{b} \mid \lambda) \propto \exp\left(-\frac{\lambda}{2}\mathbf{b}^\top \mathbf{P}\mathbf{b}\right) = \exp\left(-\frac{1}{2}\sum_{k=1}^{K-1} \lambda (b_{k+1} - b_k)^2\right). \tag{1}$$

The precision parameter λ controls the extent to which the squared differences of consecutive pairs of $\mathbf{b}$ influence the posterior, imparting a smoothing effect over the lag curve derived from $\mathbf{b}$. Since each squared difference $(b_{k+1} - b_k)^2$ contributes equally in the prior in Equation (1), variable smoothing can be achieved naturally by introducing a set of precision parameters $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_{K-1})$ that enables pairs of B-spline parameters to be smoothed more or less strongly. This generalised prior can be written $\pi(\mathbf{b} \mid \boldsymbol{\lambda}) \propto \exp\left(-\frac{1}{2}\sum_{k=1}^{K-1} \lambda_k (b_{k+1} - b_k)^2\right) = \mathrm{N}(\mathbf{0}, \mathbf{Q}^-)$ where $\mathbf{Q} = \mathbf{D}^\top \mathrm{diag}(\boldsymbol{\lambda})\mathbf{D}$, with $\mathbf{D}$ the $(K-1) \times K$ first-difference matrix, is the precision matrix

$$\mathbf{Q} = \begin{pmatrix} \lambda_1 & -\lambda_1 & 0 & 0 & \ldots & 0 \\ -\lambda_1 & \lambda_1 + \lambda_2 & -\lambda_2 & 0 & \ldots & 0 \\ 0 & -\lambda_2 & \lambda_2 + \lambda_3 & -\lambda_3 & \ldots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & -\lambda_{K-1} & \lambda_{K-1} \end{pmatrix}.$$

$\mathbf{Q}$ has rank $K-1$ and its null space is spanned by the constant vector, so the prior is intrinsic: it is improper with respect to the overall level of $\mathbf{b}$ but proper for the differences, and $\mathbf{Q}^-$ denotes a generalised inverse. The posterior distribution is nevertheless proper, since the level of $\mathbf{b}$ is identified by the likelihood. This joint specification for $\mathbf{b}$ results in the conditional distributions

$$b_k \mid \mathbf{b}_{-k}, \boldsymbol{\lambda} \sim \begin{cases} \mathrm{N}\left(b_{k+1},\ \frac{1}{\lambda_1}\right) & \text{if } k = 1 \\ \mathrm{N}\left(\frac{\lambda_{k-1} b_{k-1} + \lambda_k b_{k+1}}{\lambda_{k-1} + \lambda_k},\ \frac{1}{\lambda_{k-1} + \lambda_k}\right) & \text{if } k = 2, \ldots, K-1 \\ \mathrm{N}\left(b_{k-1},\ \frac{1}{\lambda_{K-1}}\right) & \text{if } k = K. \end{cases}$$

When $\lambda_1 = \lambda_2 = \ldots = \lambda_{K-1} = \lambda$, then $\mathbf{Q} = \lambda \mathbf{P}$, which shows that the ordinary ICAR prior in Equation (1) is a special case of the adaptive prior.

The influence of very large lags is expected to be negligible, and a lag curve that terminates away from zero at lag p would imply an artificial discontinuity with the implicitly assumed $\beta_{p+j} = 0$. The constraint $\beta_p = 0$, optionally together with $\beta_p' = 0$, is therefore imposed exactly. Writing the constraints as $\mathbf{C}\mathbf{b} = \mathbf{0}$, where the $m \times K$ matrix $\mathbf{C}$ holds the basis values and derivatives at lag p, and letting the orthonormal columns of the $K \times (K-m)$ matrix $\mathbf{Z}$ span the null space of $\mathbf{C}$, the coefficients are parameterised as $\mathbf{b} = \mathbf{Z}\tilde{\mathbf{b}}$, so that the basis becomes $\mathbf{X}\mathbf{Z}$ and the prior precision of $\tilde{\mathbf{b}}$ becomes $\tilde{\mathbf{Q}} = \mathbf{Z}^\top \mathbf{Q} \mathbf{Z}$. The null space of $\mathbf{Q}$ is spanned by the constant vector, which does not satisfy the value constraint, so $\tilde{\mathbf{Q}}$ is positive definite and the prior on $\tilde{\mathbf{b}}$ is proper. The constraint adds no tuning parameter and, as shown in Section 3, leaves the normalising constant of the prior in closed form.

A further smoothing prior distribution is placed over $\boldsymbol{\lambda}$ to restrict their flexibility. This is particularly attractive as it also reflects our intuition that the curvature of the lag function should not change too rapidly. For this purpose, an ordinary ICAR prior is used to smooth the log precisions $\boldsymbol{\tau} = (\tau_1, \ldots, \tau_{K-1})$, $\tau_k = \log \lambda_k$, such that $\boldsymbol{\tau} \mid \zeta^2 \sim \mathrm{N}(\mathbf{0}, \zeta^2 \mathbf{K}^-)$ where $\mathbf{K}$ is the $(K-1) \times$

$(K-1)$ precision matrix

$$\mathbf{K} = \begin{pmatrix} 1 & -1 & 0 & 0 & \dots & 0 \\ -1 & 2 & -1 & 0 & \dots & 0 \\ 0 & -1 & 2 & -1 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \dots & 1 \end{pmatrix}.$$

Therefore, for small values of the variance ζ^2, pairs of consecutive elements of $\boldsymbol{\tau}$ are more strongly smoothed towards each other. This random walk is itself intrinsic, with rank $K-2$, and is anchored by a prior on the level $\bar{\tau}$ of the log precisions. Both this and the prior on ζ are penalised complexity priors [Simpson et al., 2017]. The scale ζ has an exponential prior with rate chosen so that $\Pr(\zeta > 2) = 0.1$, which shrinks the model towards its non-adaptive special case unless the data ask for adaptivity. The standard deviation $\exp(-\bar{\tau}/2)$ of coefficient differences implied by the level has an exponential prior with $\Pr\{\exp(-\bar{\tau}/2) > 1\} = 0.05$ on standardised data; its density decays like $\exp(-\bar{\tau}/2)$ for large precisions, which prevents the precision from drifting without bound once the data no longer inform it, as happens wherever the true curve is exactly flat. The error variance has $\sigma^2 \sim \text{Inverse-Gamma}(0.001, 0.001)$, and unpenalised terms, an intercept and any covariates, have flat priors. Both $\mathbf{x}$ and $\mathbf{y}$ are standardised before fitting and every reported quantity is transformed back to the original scale. The combination of adaptive smoothing with smoothed precisions results in a very flexible model that can accommodate complex shapes using a relatively simple specification. The model described above assumes Gaussian errors for simplicity; autocorrelated errors are treated in Section 3, and the ideas generalise to other response distributions.

3 Computation

3.1 Integrating out the lag coefficients

A Gibbs sampler that alternates between the coefficients $\mathbf{b}$ and the log precisions $\boldsymbol{\tau}$, with Metropolis steps for the latter, is the obvious scheme for this model and was used in the first version of this paper. It mixes poorly. Conditional on $\mathbf{b}$, a large λ_k forces the difference $b_{k+1} - b_k$ to be small, and conditional on a small difference the posterior of τ_k favours an even larger precision, so the chain creeps along a funnel. The remedy is to integrate the coefficients out: given $\boldsymbol{\tau}$ and σ^2 the model is

linear and Gaussian, and the integral is available in closed form.

Let $\boldsymbol{\theta} = (\tilde{\mathbf{b}}, \boldsymbol{\gamma})$ collect the $K - m$ penalised and q unpenalised coefficients, let $\mathbf{X}$ be the corresponding $n \times (K - m + q)$ design, and define the sufficient statistics $\mathbf{S} = \mathbf{X}^\top \mathbf{X}$, $\mathbf{s} = \mathbf{X}^\top \mathbf{y}$ and $\mathbf{y}^\top \mathbf{y}$, the block-diagonal prior precision $\mathbf{Q}_\theta = \text{blockdiag}(\tilde{\mathbf{Q}}, \mathbf{0})$, $\mathbf{A} = \mathbf{Q}_\theta + \mathbf{S}/\sigma^2$ and $\mathbf{m} = \mathbf{s}/\sigma^2$. Then

$$\log p(\mathbf{y} \mid \boldsymbol{\tau}, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) + \frac{q}{2}\log 2\pi + \frac{1}{2}\log|\tilde{\mathbf{Q}}|_+ - \frac{1}{2}\log|\mathbf{A}| + \frac{1}{2}\mathbf{m}^\top \mathbf{A}^{-1}\mathbf{m} - \frac{\mathbf{y}^\top \mathbf{y}}{2\sigma^2}, \quad (2)$$

where $|\cdot|_+$ denotes the generalised determinant, the product of the non-zero eigenvalues, which is needed when $\tilde{\mathbf{Q}}$ is singular. The prior normalising constant is the only awkward term, and it has a simple form. Without end constraints $\mathbf{Q} = \mathbf{D}^\top \text{diag}(\boldsymbol{\lambda})\mathbf{D}$ is the weighted Laplacian of the chain graph on K nodes, and the matrix-tree theorem [Kirchhoff, 1847, Rue and Held, 2005] gives $|\mathbf{Q}|_+ = K \prod_{k=1}^{K-1} \lambda_k$, so that $\log|\mathbf{Q}|_+ = \log K + \sum_k \tau_k$. With a single constraint of unit normal $\mathbf{v}$, $|\tilde{\mathbf{Q}}| = |\mathbf{Q}|_+ (\mathbf{v}^\top \mathbf{1}/\sqrt{K})^2$, because $\mathbf{Z}^\top \mathbf{Q} \mathbf{Z}$ is a congruence of $\mathbf{Q}$ restricted to a hyperplane whose determinant differs from $|\mathbf{Q}|_+$ only by the squared cosine of the angle between the two hyperplane normals. In both cases $\log|\tilde{\mathbf{Q}}|$ equals $\sum_k \tau_k$ plus a constant, so every τ_k enters the normalising constant linearly; with two constraints the determinant is evaluated numerically by a Cholesky factorisation.

Evaluating Equation (2) therefore costs a single Cholesky factorisation of the $(K - m + q)$-dimensional matrix $\mathbf{A}$, of order K^3, and does not depend on n at all, since the sufficient statistics are computed once. The unpenalised coefficients are integrated out together with $\tilde{\mathbf{b}}$, which is why $\mathbf{Q}_\theta$ carries a zero block and Equation (2) carries the term $(q/2)\log 2\pi$. The derivative of Equation (2) with respect to τ_k is $\frac{1}{2}\{1 - \lambda_k \, \mathrm{E}[(b_{k+1} - b_k)^2 \mid \mathbf{y}, \boldsymbol{\tau}, \sigma^2]\}$, the expected version of the quantity that drives the conditional update, which makes clear what collapsing achieves: the expectation is taken over the coefficients instead of fixing them.

Serially correlated errors are accommodated at the same cost. If the errors follow a stationary AR(1) process with coefficient ϕ, prewhitening replaces $\mathbf{y}$ and $\mathbf{X}$ by $\mathbf{L}_\phi \mathbf{y}$ and $\mathbf{L}_\phi \mathbf{X}$, where $\mathbf{L}_\phi$ scales the first row by $\sqrt{1 - \phi^2}$ and subtracts ϕ times the previous row from every other row. The whitened sufficient statistics are quadratic in ϕ: $\mathbf{S}(\phi) = \mathbf{S}_a + \phi \mathbf{S}_b + \phi^2 \mathbf{S}_c$, with $\mathbf{S}_a = \mathbf{X}^\top \mathbf{X}$, $\mathbf{S}_b = -(\mathbf{X}_{1:}^\top \mathbf{X}_{:-1} + \mathbf{X}_{:-1}^\top \mathbf{X}_{1:})$ and $\mathbf{S}_c = \mathbf{X}_{1:-1}^\top \mathbf{X}_{1:-1}$ in terms of the design with its first or last row removed, and likewise for $\mathbf{s}(\phi)$ and $\mathbf{y}^\top \mathbf{y}$, so three sets of statistics computed once serve every value

of ϕ, and Equation (2) gains the term $\frac{1}{2}\log(1-\phi^2)$.

Finally, when the true curve is exactly flat over a range of lags, Equation (2) is flat in the corresponding log precisions beyond the point where the penalty dominates the data, and only the prior holds them down. Log precisions above 30 are treated as infinite smoothing in every term of the likelihood, which keeps $\mathbf{A}$ well conditioned without changing the posterior appreciably.

3.2 Sampling the hyperparameters

What remains is the marginal posterior of $\boldsymbol{\tau}$, σ^2, ζ and, with AR(1) errors, ϕ: a smooth density in $K+1$ or $K+2$ dimensions. It is sampled with the No-U-Turn Sampler [Hoffman and Gelman, 2014], with gradients obtained by automatic differentiation. Three details matter in practice. First, the random walk prior is parameterised non-centrally: with $w_1 = 0$, $w_{k+1} = w_k + \epsilon_k$ and $\epsilon_k \sim \mathrm{N}(0,1)$, the log precisions are $\boldsymbol{\tau} = \bar{\tau}\mathbf{1} + \zeta(\mathbf{w} - \bar{w}\mathbf{1})$, so that ζ and the increments are a priori independent and the posterior has no funnel in ζ. The scale and the level are sampled on the log scale and ϕ through its inverse hyperbolic tangent. Second, the posterior of $\boldsymbol{\tau}$ is a long ridge, since neighbouring log precisions are tied together by the random walk, and a dense mass matrix adapted during warm-up reduces the length of the trajectories several-fold. Third, the chains start at the posterior mode, found by quasi-Newton optimisation in the same parameterisation; a Gaussian approximation about that mode, with covariance from the Hessian, also serves as a fast approximate fit.

The coefficients are recovered after sampling. For each retained draw of $(\boldsymbol{\tau}, \sigma^2, \phi)$ the coefficients are drawn from their conditional distribution $\mathrm{N}(\mathbf{A}^{-1}\mathbf{m}, \mathbf{A}^{-1})$ using the Cholesky factor of $\mathbf{A}$, and the same factor gives the effective degrees of freedom of the lag curve, $\mathrm{ED} = \mathrm{tr}[(\mathbf{S} + \sigma^2\mathbf{Q}_\theta)^{-1}\mathbf{S}] - q$, the trace of the hat matrix with the unpenalised terms, which contribute exactly one degree of freedom each, removed. Draws of the lag curve $\boldsymbol{\beta} = \mathbf{B}\mathbf{Z}\tilde{\mathbf{b}}$ then yield pointwise credible bands, the cumulative effect $\sum_j \beta_j$, the lag of the largest effect and the lag beyond which the curve is negligible, all with their posterior uncertainty.

The implementation was checked in three ways. Equation (2) agrees to numerical precision with the density of $\mathbf{y}$ under its implied marginal Gaussian distribution, with and without AR(1) errors. Simulation-based calibration [Talts et al., 2018] over data sets simulated from the prior gives uniform ranks for the error variance, the mean log precision and the lag coefficients. And a Gibbs sampler with slice updates of the log precisions, validated by the joint-distribution test of

Geweke [2004], yields the same posterior as the collapsed sampler on common data.

3.3 Software

The method is implemented in the Python package `badlm` (`https://github.com/alastairrushworth/badlm`), built on JAX [Bradbury et al., 2018] and NumPyro [Phan et al., 2019]. A fit with $n = 450$, $p = 50$ and $K = 34$ takes about 20 seconds for one chain of 300 warm-up and 300 retained draws on four processor cores, and about 30 seconds for four chains of 2000 iterations; the cost grows with K rather than n. The package also provides Poisson and negative binomial responses, for which the coefficients cannot be integrated out and are sampled jointly with the log precisions; these count models sample markedly less efficiently, for the reason given at the start of Section 3.1, and a collapsed scheme for them is the subject of further work.

4 Simulation study

The adaptive model detailed in Section 2 was compared with four currently available distributed lag models, labelled $\mathcal{M}_1$, $\mathcal{M}_2$, $\mathcal{M}_4$ and $\mathcal{M}_5$, with the adaptive model as $\mathcal{M}_3$. $\mathcal{M}_1$ is the non-adaptive version of the model of Section 2, achieved by fixing $\lambda_1 = \lambda_2 = \ldots = \lambda_{K-1} = \lambda$. $\mathcal{M}_2$ is a Bayesian implementation of the DLM of Muggeo [2008], in which the basis coefficients are shrunk by a ridge penalty that increases linearly with lag, $\mathbf{Q} = \lambda\,\mathrm{diag}(1, 2, \ldots, K)$. $\mathcal{M}_4$ is the P-spline model of Obermeier et al. [2015], with the first-difference penalty of $\mathcal{M}_1$ plus a ridge penalty, $\mathbf{Q} = \lambda\mathbf{P} + \rho\mathbf{I}$. $\mathcal{M}_5$ is the DLM with logarithmically spaced knots of Gasparrini et al. [2010], fitted by least squares with the number of interior knots, between zero and twelve, chosen by AIC. Models $\mathcal{M}_1$ to $\mathcal{M}_4$ share the same basis of $K = \lceil 2(p+1)/3 \rceil$ uniformly spaced cubic B-splines with the curve constrained to zero at the maximum lag, so that they differ only in the penalty; their scale parameters λ and ρ carry the penalised complexity prior of Section 2 that the adaptive model places on the level of its log precisions. Every Bayesian model was fitted with the collapsed sampler of Section 3, with one chain of 300 retained draws after 300 warm-up iterations started at the posterior mode, and the posterior mean of the lag curve was taken as the estimate. Effective degrees of freedom are those of the lag curve, and for $\mathcal{M}_5$ the number of basis functions selected. The study is reproduced by a script in the software repository.

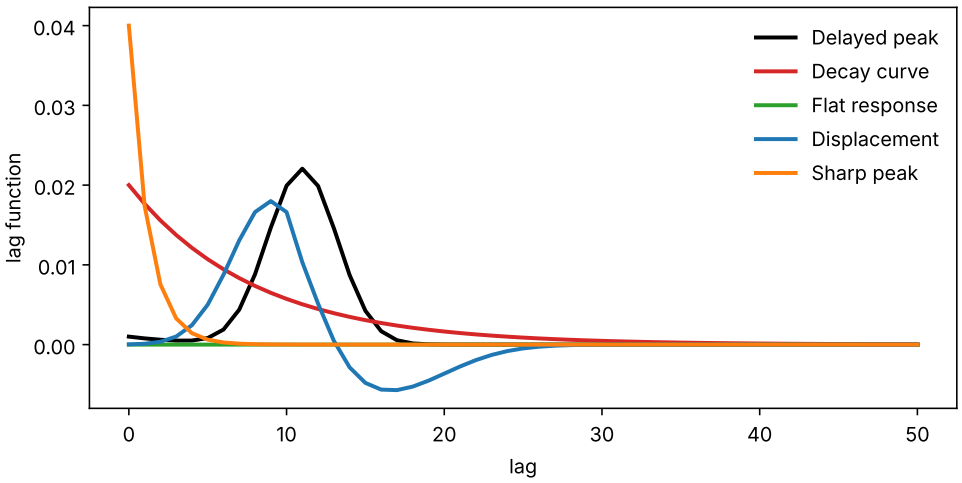


Figure 1: The five lag curves used to simulate response and covariate data with different lag response.

4.1 Comparison under different lag structures

Time series response and covariate data were simulated under each of five lag relationships with maximum lag $p = 50$, shown in Figure 1: a delayed peak scenario (DELAYED PEAK), an exponential decay curve (DECAY CURVE), a horizontal line at zero equivalent to a null effect (FLAT RESPONSE), a curve exhibiting a shape similar to a mortality displacement effect (DISPLACEMENT) and a sharply peaked function that declines rapidly to zero (SHARP PEAK). A covariate series was generated as $x_t \sim \mathrm{N}(0.5x_{t-1}, 0.1^2)$ and standardised, and conditional on each lag function a response series was generated as $y_t = \sum_{j=0}^{50} \beta_j x_{t-j} + \epsilon_t$ for $t = 51, \ldots, 500$, with autocorrelated errors $\epsilon_t \sim \mathrm{N}(0.2\epsilon_{t-1}, 0.1^2)$. The lag function explains about 30 per cent of the variance of y_t under the DELAYED PEAK curve. The autocorrelation in both x_t and ϵ_t reflects the temporal dependence in covariates and errors typically observed in real data.

Models $\mathcal{M}_1$ to $\mathcal{M}_5$ were fitted to each of 50 realisations of the processes $\{x_t, y_t\}$ described above. The models were assessed in terms of their ability to recover the true lag curve, as measured by root mean squared error (RMSE) and squared bias (Bias^2) of the estimates, defined for the jth simulation as $\left(\frac{1}{51}\sum_{i=0}^{50}(\hat{\beta}_i^{(j)} - \beta_i)^2\right)^{1/2}$ and $\left(\frac{1}{51}\sum_{i=0}^{50}(\hat{\beta}_i^{(j)} - \beta_i)\right)^2$, and, following Hodges and Sargent [2001], by the effective degrees of freedom of the fitted curve.

Table 1 summarises the results. The adaptive model $\mathcal{M}_3$ has the lowest RMSE under each of the four lag curves with a non-null effect, and the margin is widest where the curve changes

Lag curve shape	Summary	$\mathcal{M}_1$	$\mathcal{M}_2$	$\mathcal{M}_3$	$\mathcal{M}_4$	$\mathcal{M}_5$
	RMSE	1.98	2.26	1.53	2.12	2.54
Delayed peak	Bias^2	0.50	0.48	0.40	0.52	0.52
	ED	13.32	17.86	8.84	15.48	12.32
	RMSE	1.45	1.99	1.29	1.51	1.45
Decay curve	Bias^2	0.48	0.53	0.43	0.50	0.52
	ED	8.80	15.32	7.04	9.38	5.28
	RMSE	0.59	0.62	0.59	0.70	1.35
Flat response	Bias^2	0.32	0.08	0.32	0.17	0.51
	ED	2.57	4.29	2.52	4.63	4.98
	RMSE	1.95	2.14	1.61	2.11	2.36
Displacement	Bias^2	0.50	0.37	0.42	0.48	0.51
	ED	13.02	16.58	9.29	15.33	10.94
	RMSE	2.84	2.23	1.29	2.92	1.69
Sharp peak	Bias^2	0.54	0.35	0.35	0.55	0.51
	ED	14.00	15.08	5.65	16.06	6.56

Table 1: Estimated lag function RMSE, squared bias and effective degrees of freedom for five types of lag curve shape, averaged across 50 simulated data sets. RMSE has been scaled by 10^3 and Bias^2 by 10^6.

character along the lag axis. Relative to the best of the other four models its RMSE is 23 per cent lower under the Delayed peak curve, 24 per cent lower under Sharp peak, 17 per cent lower under Displacement and 11 per cent lower under Decay curve, and in each of these scenarios it attains the lower error in at least 96 per cent of the 50 replicates when compared replicate by replicate with $\mathcal{M}_1$, $\mathcal{M}_2$ or $\mathcal{M}_4$. The single-penalty P-spline $\mathcal{M}_1$ is the strongest of the other Bayesian models under the three smooth curves, but under Sharp peak, where one smoothing parameter must compromise between the steep rise at short lags and the long flat tail, it and $\mathcal{M}_4$ are the worst models of all, with RMSE more than double that of $\mathcal{M}_3$. The ridge component of $\mathcal{M}_4$ adds a little error to $\mathcal{M}_1$ in every scenario, and the lag-increasing ridge of $\mathcal{M}_2$ does so under the three smooth curves, although it helps under Sharp peak, whose effect is confined to the short lags that its penalty leaves relatively free. The log-knot model $\mathcal{M}_5$ is competitive under Decay curve and Sharp peak, whose features lie at the short lags where its knots are dense, but it has the largest error of all under Delayed peak, whose feature lies at intermediate lags, and under Flat response, where AIC selects spurious structure. Under that null curve the four Bayesian models are close, and $\mathcal{M}_1$ and $\mathcal{M}_3$ are indistinguishable, each using about 2.5 effective degrees of freedom: the adaptive prior does not invent structure when there is none to find. Under the four

non-null curves the squared bias is a small part of the mean squared error for every model, so the differences in RMSE reflect the variance of the estimates rather than systematic error in their level; under FLAT RESPONSE, where the fitted curves are almost straight lines, the error of a Bayesian model is mostly a small shift in the level of the curve, which the ridge penalties of $\mathcal{M}_2$ and $\mathcal{M}_4$ shrink towards zero.

The effective degrees of freedom show the mechanism. $\mathcal{M}_3$ uses fewer effective parameters than $\mathcal{M}_1$, $\mathcal{M}_2$ and $\mathcal{M}_4$ under every curve, and under the four non-null curves it uses between a fifth and three fifths fewer than $\mathcal{M}_1$ (for example 5.7 against 14.0 under SHARP PEAK and 8.8 against 13.3 under DELAYED PEAK). Lower error with fewer effective parameters is the signature of adaptive smoothing: the lag-varying precision is high, and the fitted curve close to linear, over the flat part of each curve, and is relaxed only where the curve actually moves. The price is computational. With the collapsed sampler of Section 3 a fit of $\mathcal{M}_3$ took about 25 seconds on a single CPU core against 10 to 13 seconds for $\mathcal{M}_1$, $\mathcal{M}_2$ and $\mathcal{M}_4$, whose hyperparameter spaces have one or two dimensions rather than $K-1$, and well under a second for the least squares fit of $\mathcal{M}_5$.

4.2 DLM misspecification with maximum lag p

Next, simulation was used to explore the robustness of $\mathcal{M}_1$ to $\mathcal{M}_5$ to different choices of maximum lag p. The series x_t and y_t were generated exactly as in Section 4.1, using only the DISPLACEMENT lag function with 50 lags, and the models were fitted assuming maximum lags of $p = 50, 75, 100$ and 125, with $K = \lceil 2(p+1)/3 \rceil$ basis functions so that the flexibility available to each model grows in proportion to p. For each of 20 realisations the RMSE of the lag curve estimate over all $p+1$ lags, its squared bias and the effective degrees of freedom were recorded; Figure 2 summarises the results.

Figure 2 shows that the adaptive model is almost indifferent to the choice of maximum lag. Its effective degrees of freedom rise only from 9.1 at $p = 50$ to 10.9 at $p = 125$, although the number of basis functions, and so the flexibility on offer, grows from 34 to 84: the random walk prior on the log precisions settles on a high precision across the added lags, where the data show nothing, and the fitted curve there is a straight line at zero. Its RMSE, which averages the squared error over all $p+1$ lags, consequently falls from 1.40 to 1.11, because the added lags are estimated almost without error and dilute the error in the first 50, and $\mathcal{M}_3$ has the lowest error of the five models at

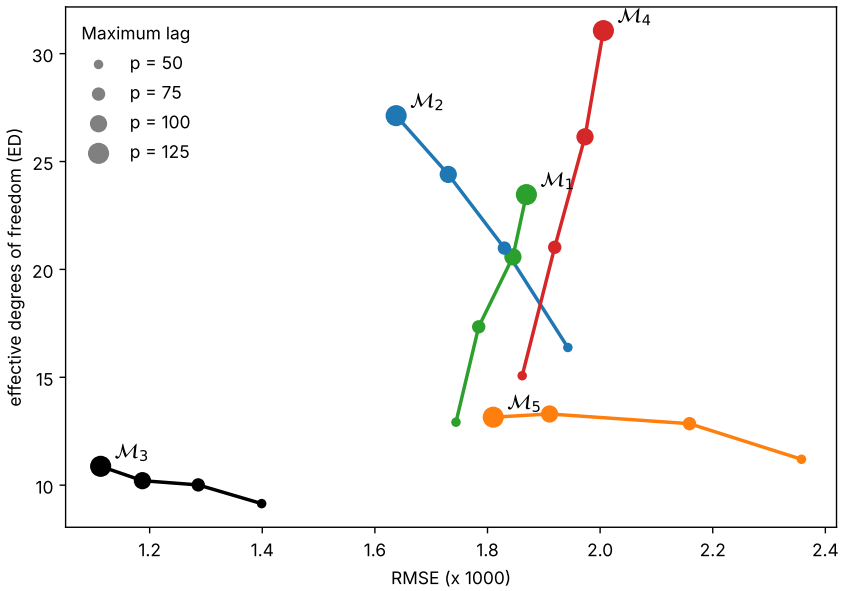


Figure 2: Effective degrees of freedom and RMSE for each of $\mathcal{M}_1$ to $\mathcal{M}_5$. Increasing diameter of the plotting symbols indicates larger numbers of maximum lags assumed in advance of model fitting.

every p, in every one of the 20 replicates when compared with $\mathcal{M}_1$, $\mathcal{M}_2$ or $\mathcal{M}_4$. The single-penalty P-spline $\mathcal{M}_1$ and the P-spline with ridge $\mathcal{M}_4$ behave quite differently: their effective degrees of freedom grow almost in proportion to p, from 12.9 to 23.5 and from 15.1 to 31.1, because the one smoothing parameter must stay small enough to follow the DISPLACEMENT shape over the first 50 lags and then lets the curve wander over the lags that follow, and their RMSE rises with p even though the added lags should be easy to estimate. The lag-increasing ridge of $\mathcal{M}_2$ does shrink the distant lags, so its RMSE falls with p, but its effective degrees of freedom still climb to 27 and its error at $p = 125$ remains about 50 per cent above that of $\mathcal{M}_3$. The log-knot model $\mathcal{M}_5$ keeps its degrees of freedom roughly constant by construction, because its knots are concentrated at short lags, but it is the least accurate model of all at $p = 50$ and 75. The cost of the adaptive model's $K - 1$ hyperparameters is again computational: its fitting time grows from 28 seconds at $p = 50$ to about two minutes at $p = 125$, against 11 to 19 seconds for the other Bayesian models at every p.

5 Illustrations with real data

Four applications illustrate how the adaptive lag curve behaves on noisy real data. All use the accompanying software with its default settings: the random walk prior on the log precisions, an exact zero constraint at the maximum lag, $K = \lceil 2(p+1)/3 \rceil$ basis functions, and the collapsed sampler of Section 3 with two chains of 1000 retained draws. Effective degrees of freedom are reported for the lag curve alone, excluding unpenalised terms. The scripts that reproduce them are in the software repository.

5.1 Daily stock market data

Prices are not stationary, so they are transformed to percentage log returns $r_t = 100 \log(P_t/P_{t-1})$. Daily closing prices of the S&P 500 index and of Apple, and closing values of the VIX implied volatility index, were obtained for 2007 to 2016, 2,306 trading days spanning the 2008 crisis. Figure 3 shows four lag curves. Panels (a) and (b) regress the next day's return on the preceding 20 days' returns, the linear information that a trading rule could exploit. For the index the curve is negative at lag one, -0.10 (95% interval -0.15 to -0.06) per percentage point, and returns to zero within a week: a short-lived reversal with cumulative effect -0.23 (-0.42 to -0.03) and 4.7 effective degrees of freedom. For Apple the curve is indistinguishable from zero at every lag, with 1.7 effective degrees of freedom; the adaptive smoother reports an absence of structure rather than inventing it. Panel (c) regresses the daily change in log VIX on the same day's and the preceding 60 days' index returns. The contemporaneous coefficient is -4.1 (-4.2 to -3.9), so implied volatility rises about 4 per cent for each percentage point the index falls; it is followed by a small positive rebound over the following week as the shock partly reverses, and by nothing beyond about ten days. Panel (d) regresses the log absolute return, a measure of realised volatility, on its own past over 100 days. The curve is strongest at lags two to three, decays over about a month and remains small but mostly positive thereafter, the long memory of volatility; the cumulative elasticity is 0.84 (0.75 to 0.94) and the smoother describes 100 lags with 7.9 effective degrees of freedom. In each panel the band widens where the data are least informative and the curve is pinned to zero at the maximum lag.

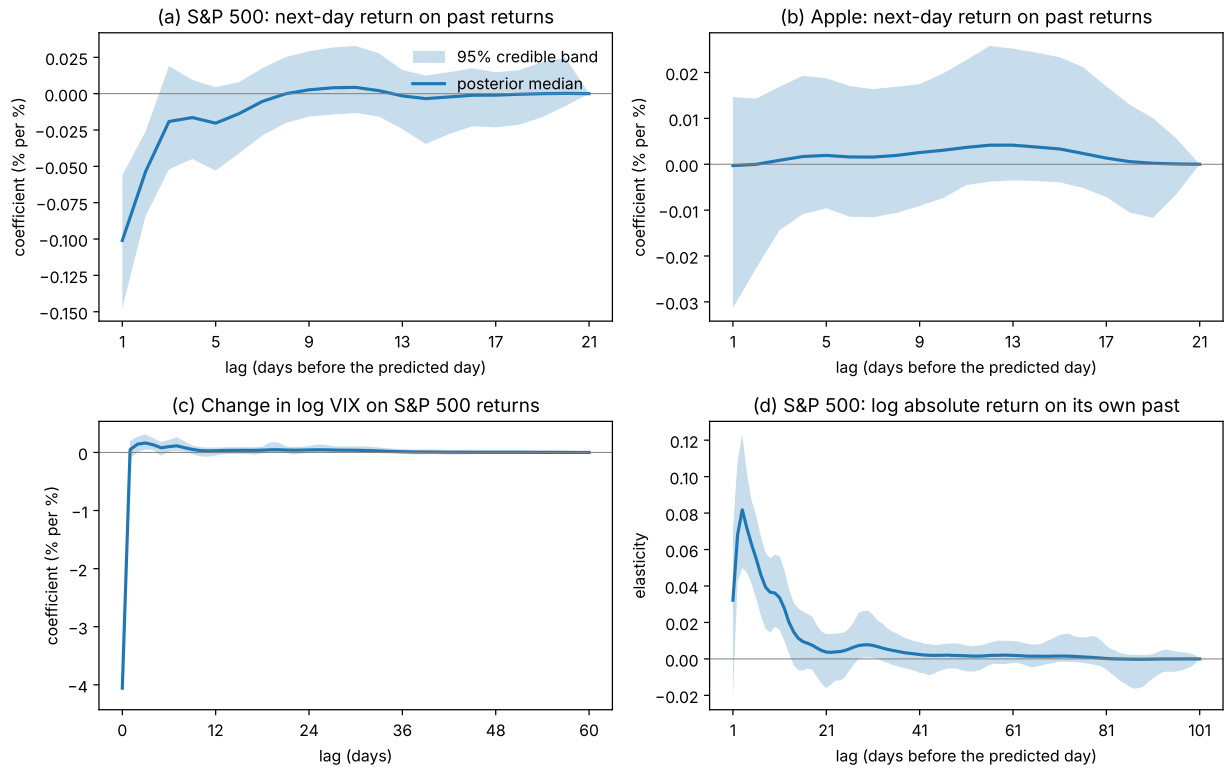


Figure 3: Lag curves for daily stock market data, 2007 to 2016: posterior medians with 95% credible bands. (a) Next-day S&P 500 return on the previous 20 days' returns; (b) the same for Apple; (c) daily change in log VIX on contemporaneous and lagged S&P 500 returns; (d) log absolute S&P 500 return on its own past over 100 days.

5.2 Intraday order flow and prices

At the other extreme of time scale, the sample data distributed with the QuantConnect Lean repository contain every trade and quote for the exchange traded fund SPY and for the stock BAC on the five trading days from 7 to 11 October 2013. They were reduced to one-second series: the midquote from the best bid and offer of the four most active venues, its return in basis points, the order-flow imbalance (the number of buyer-initiated minus seller-initiated trades, signed against the prevailing midquote), volume and the quoted spread. Lags were formed within each day, so that the first second of a day is never regressed on the previous afternoon, and B-splines of time of day entered as unpenalised terms to absorb the intraday pattern in volume and volatility. Each series has 117,000 seconds, and all fits took the default settings. Figure 4 shows the results.

Panel (a) regresses the midquote return on the order-flow imbalance of the same second and the 120 before it, and panel (b) shows the cumulative sum of these coefficients, the response of the price

to one unit of order flow as a function of time since the trade, with its credible band. One more buy than sell trade moves the midquote by 0.0061 basis points in the same second; the response continues to build for about seven seconds, to 0.0093, and about 15 per cent of it then decays away over the following minute, leaving a permanent impact of 0.0079 (0.0074 to 0.0084). This is the shape that microstructure models describe with a transient propagator [Hasbrouck, 1991, Bouchaud et al., 2004], estimated here without assuming its functional form. Panels (c) and (f) regress the imbalance on its own past over 300 seconds for SPY and BAC. The coefficient at one second is 0.16 for SPY and 0.11 for BAC, but the tails carry most of the memory: the coefficients over five minutes sum to 0.38 and 0.57 respectively, and the curve for BAC, a large-tick stock whose queue dynamics encourage order splitting, remains positive to about 100 seconds [Lillo and Farmer, 2004]. Panels (d) and (e) show the two directions of the relation between volume and volatility: a thousand shares traded a second earlier raise the next second's absolute return by 0.001 basis points, with a decay over about a minute and a secondary bump at six to seven seconds that coincides with the time taken to absorb order flow in panel (b); and one basis point of absolute return attracts about 1,400 additional shares in the following second and some 22,000 over the following two minutes. The six curves use between 8 and 12 effective degrees of freedom for 120 to 300 lags.

5.3 Air pollution and mortality in Chicago

The setting that motivated much of the distributed lag literature is the delayed effect of air pollution on daily mortality, and in particular the question of mortality displacement raised in Section 1. The `chicagoNMMAPS` data distributed with the R package `dlnm` [Gasparrini, 2011] contain daily counts of all-cause, cardiovascular and respiratory deaths in Chicago from 1987 to 2000, 5,114 days with means of 115, 51 and 9 deaths per day, together with PM10 (mean 34 micrograms per cubic metre, 5 per cent of days interpolated), temperature and dew point. Log deaths were regressed on 40 lags of PM10 with the usual adjustments of this literature as unpenalised terms: a B-spline of time with seven degrees of freedom per year for trend and season, weekday indicators, and B-splines of temperature and dew point, with AR(1) errors. Coefficients are expressed as the percentage change in deaths per 10 micrograms per cubic metre.

Figure 5 shows the three lag curves and their cumulative sums. For all-cause deaths the curve is positive on the day of exposure, 0.15 per cent (95% interval -0.02 to 0.36), falls through zero

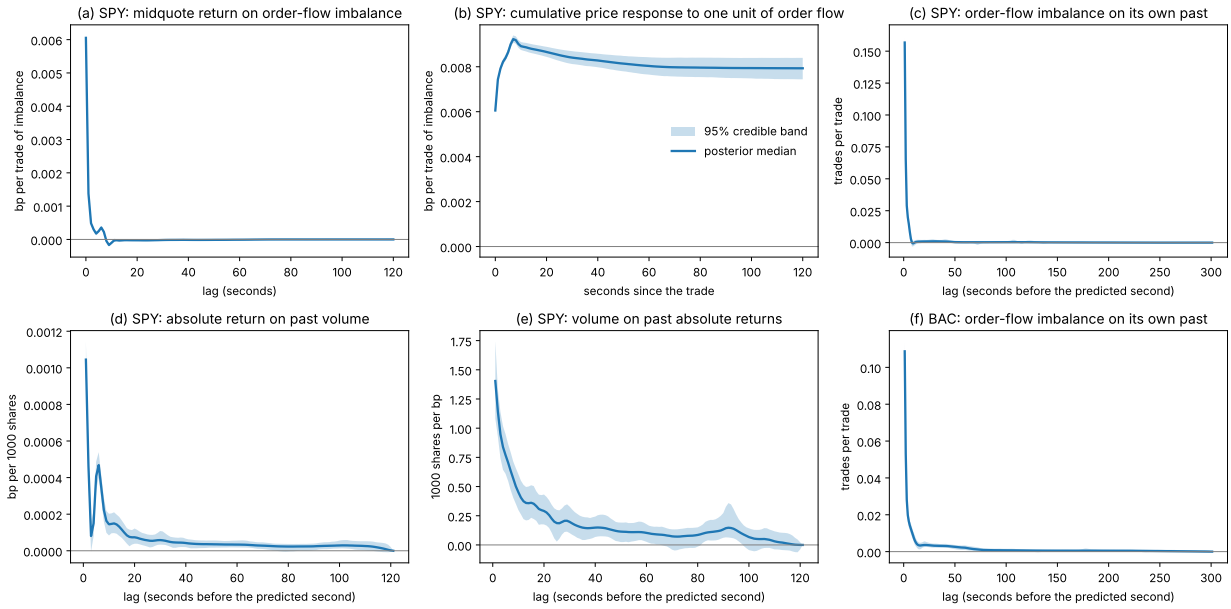


Figure 4: Intraday lag curves at one-second resolution, 7 to 11 October 2013. (a) SPY midquote return on order-flow imbalance; (b) the cumulative sum of (a), the price response to one unit of order flow; (c) SPY order-flow imbalance on its own past; (d) SPY absolute return on past volume; (e) SPY volume on past absolute returns; (f) BAC order-flow imbalance on its own past.

within three days and is negative from about lag four to lag ten, where the band excludes zero, before returning to zero. The cumulative effect over 40 days is −0.20 per cent (−1.30 to 0.85): deaths are brought forward by a few days rather than added, the displacement pattern of Schwartz [2000] and Zanobetti et al. [2000], and the lag curve exhibits it without any lag structure having been specified in advance. The curve uses about five effective degrees of freedom. Cardiovascular deaths show the same shape with smaller amplitude, and respiratory deaths, at nine per day, are too few to show anything beyond a wide band around zero.

5.4 COVID-19 cases and deaths

Deaths from COVID-19 follow infections after a delay of a few weeks, so the lag curve relating daily deaths to earlier cases estimates both the size of the relationship and the shape of the delay. Daily counts for the United States from June 2020 to March 2021, 304 days with a mean of 1,471 deaths per day, were taken from the compilation by The New York Times; the period precedes any effect of vaccination on the case fatality ratio. The exposure is the log of the seven-day trailing mean of new cases and the response is log deaths, so the lag coefficients are elasticities. Weekday indicators

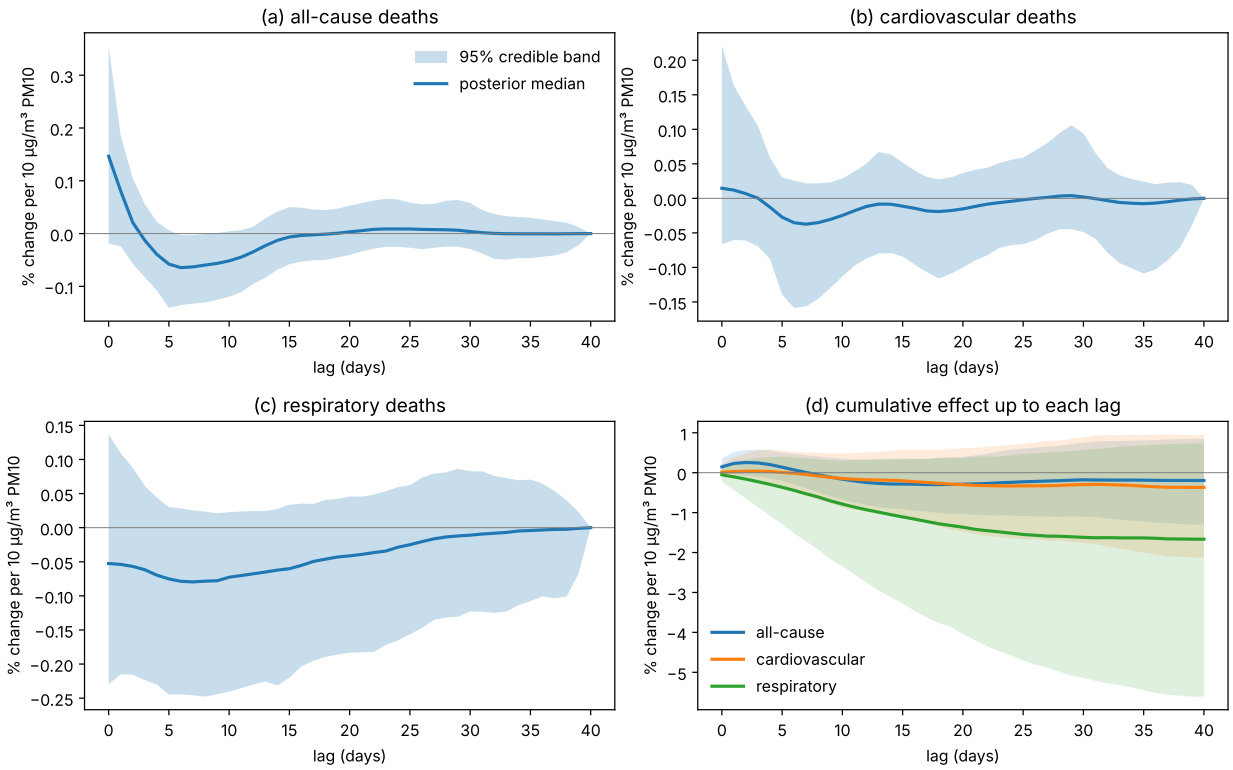


Figure 5: Chicago, 1987 to 2000: percentage change in daily deaths per 10 micrograms per cubic metre of PM10 at lags 0 to 40 for (a) all-cause, (b) cardiovascular and (c) respiratory deaths, with 95% bands, and (d) the cumulative effect up to each lag.

absorb the day-of-week pattern in death reporting and the errors follow an AR(1) process to allow for the remaining serial correlation. Figure 6 shows the data and the fitted curve with $p = 35$. The cumulative elasticity is 0.74 (0.66 to 0.81): a 10 per cent rise in cases is followed by roughly a 7 per cent rise in deaths, the shortfall from one reflecting the fall in the case fatality ratio over the period. The curve rises to a peak at about three weeks and returns to zero by five, using under three effective degrees of freedom, but the individual lag coefficients are poorly identified, because the smoothed case series is highly collinear across lags: the data determine the total and the timing, and the adaptive prior supplies the smoothness. Deaths reported on Tuesday to Friday are about 2.5 times those reported on a Sunday. A negative binomial model for the raw counts, shown dashed in Figure 6, agrees on the total elasticity, 0.73 (0.68 to 0.77), and on the peak, but its sampler did not converge satisfactorily, with many divergent transitions and $\hat{R}$ of 1.4: the dependence between the curve and its smoothing parameters that the Gaussian model integrates out remains present in a count model, and a collapsed or approximately collapsed scheme for count responses is a topic

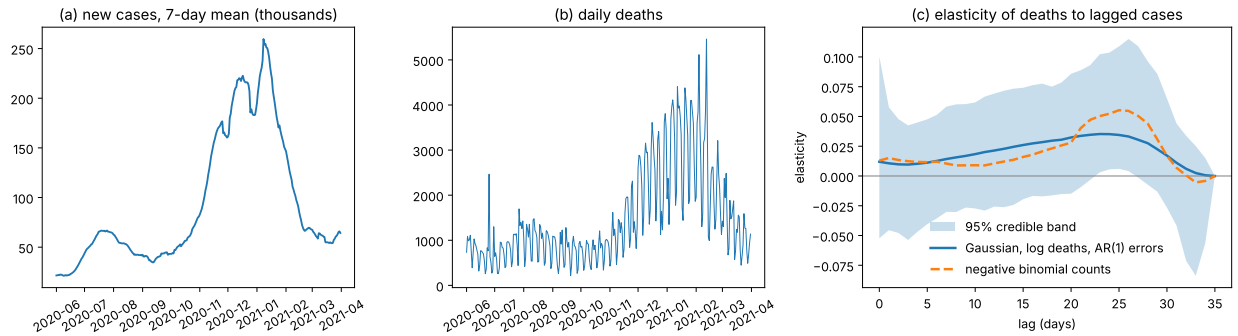


Figure 6: COVID-19 in the United States, June 2020 to March 2021. (a) Seven-day mean of new cases; (b) daily deaths; (c) elasticity of deaths to cases at lags 0 to 35 under the Gaussian model for log deaths with AR(1) errors (posterior median and 95% band) and under a negative binomial model for the counts (dashed).

for further work.

6 Discussion

This paper makes several important contributions. It was shown by simulation in Section 4 that estimation of lag structure can strongly depend on the type of smoothing model that is assumed. Several existing DLM models were shown to be non-robust to the choice of maximum lag p, even when the underlying lag function is identical, which suggests that the interpretation of lag estimates should be made with caution. A new model has been proposed that combines automatic adaptive smoothing with a pragmatically large choice of p to ensure simple and flexible smoothing of the lag curve that avoids sensitivity to the choice of p, together with an inference scheme in which the lag coefficients are integrated out analytically, so that the remaining posterior is low dimensional, free of the funnel that hampers a Gibbs sampler, and sampled at a cost that does not grow with the length of the series. The real data illustrations in Section 5 show the adaptive curve reporting no structure where the data contain none, a sharp and short-lived response where there is one, a slowly decaying response over a hundred lags, the build-up and partial decay of price impact over seconds, and the displacement of deaths by a few days after a polluted day, in each case with far fewer effective parameters than the number of lags. The new approach provides users of DLMs with a new way to explore their data with confidence that the estimates are not contaminated by artefacts that result from particular model choices.

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